

Tide retrospective:

Bounded-prior quartic partition function (grammar precursor)

Daniel Murfet (with Claude Opus 4.7)

2 May 2026

Abstract

A single Tide-loop excursion in the `laplace` seabed, formalising the smallest possible step toward the grammar paper’s $Z_n[\varphi] = \int \varphi(w)e^{-nK(w)}\pi(w) dw$ machinery: replace the integration domain $\mathbb{R}$ in the existing partition function by a bounded interval $[-a, a]$ and quantify the resulting boundary correction.¹ The proof adds 232 lines of Lean to the seabed. Zero `sorry`, zero `axiom`, zero `native_decide`; clean diagnostics on first build after a parser fix on the absolute-value statement. The interesting moves were upstream: a deliberation step that took an over-ambitious direction (the full 2D $K = w_1^2 w_2^4$ resolution example, RLCT 3/4 with logarithm) and pushed it down to a 1D one-ingredient-at-a-time precursor, and a numerical sanity check that surfaced a loose exponent constant in the candidate statement before any Lean was written.

1 Setting

This Tide step sits in the same `laplace` seabed used by five preceding tide steps (the quartic moments, 2D semi-degenerate, universal scaling, f_1 large- t , and 2D pure-quartic excursions),² but it is the first tide aimed not at the Susceptibility Primer’s Section 4 but at the *grammar paper* (Gerraty et al., *Expectations and the Exceptional Divisor*). Per the cross-tide candidates survey,³ the grammar paper’s *simplest resolution example* — $K(w) = w_1^2 \cdot w_2^4$ on a 2D parameter space with bounded prior, leading asymptotic $Z_n[\varphi] \sim C_\varphi \cdot n^{-3/4} \log n$ with RLCT $\lambda = 3/4$ and multiplicity $m = 2$ — is the natural next target, but the survey itself flagged the full 2D form as “probably the wrong shape for a single excursion” and asked Step 2 to pick a 1D precursor instead.

The Step 2 deliberation drafted three candidates of varying ambition: (A) bounded-prior quartic partition function with explicit exponential remainder; (B) the same with a harmonic potential (essentially glue-of-existing-seabed-lemmas); (C) bounded-prior quartic with a continuous test function (the genuine 1D Watanabe asymptotic at a regular critical point). GPT-5.5 Pro and Claude both backed A, on matched grounds: B is too small to count as a step outward (the harmonic tail is already in the seabed and there is no new quartic-side analysis to do), C is too large because it bundles bounded-prior, test-function localisation, change-of-variables, and dominated convergence in one step. A introduces *exactly one* ingredient (bounded prior, with the necessary new quartic tail bound) on the most-recently-finished tide step’s potential. The full deliberation, with the GPT consult preserved verbatim alongside, lives in the grammar project’s tide-log.⁴

¹Headline theorem: `Laplace.OneD.quartic.partition.bounded.prior`.

²Branches: `tide/quartic-moments`, `tide/2d-semi-degenerate`, `tide/universal-scaling`, `tide/f1-large-t`, `tide/2d-pure-quartic`.

³`projects/automation/log/2026-05-02-tide-candidates-survey.md`

⁴`projects/grammar/tide-log/2026-05-02-tide-grammar-precursor.md` and `gpt55.tide.grammar.v1.md` alongside it.

The branch is chained on top of the prior 2D pure-quartic tide’s tip⁵ per the linear-chain branch-hygiene rule, so a single SHA-pin covers all in-flight tides on the way to a squash-merge.

2 The thing formalised

For the centred quartic potential

$$L(w) = \frac{w^4}{24}$$

and any $a > 0$, $t > 0$, the bounded-prior partition function differs from the existing full-line `quartic.partition` by an exponentially small correction:

$$\left| \int_{-a}^a e^{-t \cdot w^4 / 24} dw - \frac{1}{2} (24/t)^{1/4} \Gamma(1/4) \right| \leq \frac{24}{t \cdot a^3} \exp\left(-\frac{t \cdot a^4}{24}\right).$$

The Lean signature:

```
theorem quartic_partition_bounded_prior {t a : ℝ} (ht : 0 < t) (ha : 0 < a) :
  |(w in Icc (-a) a, Real.exp (-(t * w ^ 4 / 24))) -
   (1/2) * (24/t) ^ ((1 : ℝ) / 4) * Real.Gamma ((1 : ℝ) / 4)| ≤
  (24 / (t * a ^ 3)) * Real.exp (-(t * a ^ 4 / 24))
```

located at `Laplace/OneD/QuarticBoundedPrior.lean`, in namespace `Laplace.OneD`.

This is the simplest possible step toward the grammar paper’s $Z_n[\varphi]$ machinery. The grammar paper’s partition function differs from the existing seabed in three ways: (i) bounded prior π replacing the full real line, (ii) test function φ inside the integrand, (iii) explicit asymptotic in n with remainder. The present theorem adds *only* (i), on the most degenerate 1D potential we already have, with a remainder that is not just asymptotic but *exact* for all $t > 0$. Subsequent tide steps can layer (ii) and the monomial generalisation on top, with the bounded-prior tail-bound infrastructure already in place.

Numerical sanity check: `scipy.integrate.quad` on $[-a, a]$ for $(a, t) \in \{1, 2\} \times \{10, 100, 1000\}$ agreed with leading + tail bound to machine precision (e.g. at $a = 1, t = 100$: numeric 1.2672, leading 1.2688, error 1.6×10^{-3} , tail bound 3.7×10^{-3}). The same script also validated candidate B (harmonic) and candidate C (cosine test function, ratio $\rightarrow 1$ slowly as $t \rightarrow \infty$); recorded in the tide log.

3 Proof strategy

The proof has three pieces: a tail sublemma, a tail-symmetry lemma, and a splitting identity that reduces the headline to combining them.

The *key sublemma* is `quartic_tail_loi`:

$$\int_a^\infty e^{-t \cdot w^4 / 24} dw \leq \frac{12}{t \cdot a^3} \exp\left(-\frac{t \cdot a^4}{24}\right).$$

The proof reduces the quartic tail to the existing Gaussian tail. For $w \geq a$ we have $w^2 \geq a^2$, hence $w^4 = w^2 \cdot w^2 \geq a^2 \cdot w^2$, hence

$$e^{-(t/24)w^4} \leq e^{-(ta^2/24)w^2} = e^{-(b/2)w^2} \quad \text{with } b = ta^2/12.$$

⁵Branch `tide/grammar-precursor-bounded-prior` on top of `tide/2d-pure-quartic` at commit `efafbfa`.

Apply the seabed’s `Laplace.OneD.gaussian.tail_bound.rescaled.Ioi` at $M = a$, $b = ta^2/12$:

$$\int_a^\infty e^{-(bw^2)/2} dw \leq \frac{e^{-ba^2/2}}{b \cdot a} = \frac{12}{ta^3} \exp\left(-\frac{ta^4}{24}\right).$$

That bound is already what we want. The Lean form glues a set-integral monotonicity lemma on the pointwise bound to the seabed’s Gaussian tail bound,⁶ with integrability of the quartic side from the seabed’s `quartic.integrable.pow` and of the Gaussian comparator from Mathlib’s `integrable.exp_neg_mul_sq`.

The *tail-symmetry lemma* mirrors the seabed’s Gaussian-side analogue, using Mathlib’s negation-substitution lemma together with the evenness identity $(-w)^4 = w^4$.

The *splitting identity* writes $\mathbb{R}$ as `Iic a` $\cup$ `Ioi a` via Mathlib’s `integral.Iic_add_Ioi`, then `Iic a` as `Iio (-a)` $\cup$ `Icc (-a, a)` via `setIntegral.union` with a hand-written disjointness witness and a six-line `ext/rcases` union proof.⁷ Combined with the tail symmetry, this gives

$$\int_{\mathbb{R}} e^{-tL} = \int_{[-a,a]} e^{-tL} + 2 \int_{(a,\infty)} e^{-tL}.$$

The *headline* then unfolds in eight lines: the full-line integral equals `quartic.partition` (via `quartic.moment_even` at $n = 0$, after a small `Nat.cast_zero / mul_zero / zero_add` cleanup); the absolute-value bound on the difference equals $2 \cdot \int_{(a,\infty)}$, which by `quartic.tail.Ioi` doubled is $(24/(ta^3)) \cdot \exp(-ta^4/24)$.

The total weight of the proof is concentrated in the `setIntegral.mono.on` step (where the integrability bookkeeping costs more than the mathematical bound itself) and the union-and-disjointness glue for `setIntegral.union`. Both are mechanical once the strategy is right.

4 Roadblocks and resolutions

The deliberation surfaced two issues before any Lean was written, and the formalisation surfaced two more.

Loose exponent in the original draft. Claude’s first version of candidate A had the remainder bound $|r_a(t)| \leq C_a \cdot \exp(-ta^4/48)$, derived from a square-completion comparison $w^4/24 - w^2 \geq -6/t$. The factor 48 is correct but loose by a factor of 2 in the exponent and folds the t into an unstructured constant C_a . The numerical sanity check immediately showed the bound was $\sim 100\times$ slack at $a = 1, t = 100$, and GPT-5.5 Pro independently flagged the cleaner alternative:

GPT-5.5 Pro: A *cleaner and sharper* bound is available: $|r_a(t)| = 2 \int_a^\infty e^{-tw^4/24} dw \leq \frac{24}{ta^3} e^{-ta^4/24}$. This comes from the simple comparison $w^4 \geq a^2 w^2$ for $w \geq a$, reducing the quartic tail to the already-formalised Gaussian tail.

That same paragraph turned into the proof strategy. Calibration: the numerical check sanity-checked both the leading constant and the remainder shape, and the deliberation step caught the tightenable exponent before any Lean compilation. Two reviewer cycles saved.

⁶`setIntegral.mono.on`, applied to `gaussian.tail_bound.rescaled.Ioi`.

⁷The new module exports this as `quartic.partition.split`.

Disagreement on candidate scope. Both parties backed A, but GPT also flagged a Lipschitz-near-zero middle-ground candidate (essentially candidate C with weakened hypothesis on φ) as worth considering. We deferred: the rule “one ingredient at a time” wins over “cleanest single statement that combines two ingredients.” A future tide can pick up the test-function side from this base.

Lean parser strict on `| placement`. The first version of the headline statement was written `|(( ...) - leading) | ≤ ...` with a space before the closing `|`. Lean’s parser failed at column 67 with “expected no space before”. The fix was to drop the redundant outer parens: `|(...) - leading| ≤ ....` The lesson is that the absolute-value notation is more sensitive to surrounding whitespace than other delimiter-like operators in Lean; in doubt, keep the inside terse.

Concurrent-agent staging race (recurrence). The repository hosting this Tide step had two other Tide branches running concurrently (the 2D pure-quartic and sextic-moments excursions).⁸ The top-level import file `Laplace.lean` was rewritten by the parallel agents twice during the run, each time stripping or reordering imports to match what was on *their* branch’s HEAD. The fix in this run was to (a) stash the working tree, (b) check out the import file from `main` to a known baseline, (c) re-apply only the imports relevant to this branch and the prior chained-base branch, then (d) force-retarget the branch label onto the prior tide’s tip before committing.

This is the same staging race called out in the 2D pure-quartic retrospective. The structural fix is independent of either tide: concurrent agents need filesystem isolation via `git worktree`. That’s repeated under *Lessons* below.

Beyond these four, no proof-side roadblocks. All four sub-theorems typechecked clean within ten minutes once the strategy was pinned.

5 What was Mathlib, what was new

Mathlib provided. The Gaussian integrability lemma for the comparator, the set-integral monotonicity lemma for pulling a pointwise inequality through, the two splitting lemmas (`Iic ∪ Ioi` and the `setIntegralUnion` disjoint-set decomposition), the negation substitution `integralCompNegIic` for the tail symmetry, plus the standard `Real.expLeExp`, `mulSelfLeMulSelf`, `powPos`, and the `ring/fieldSimp/linarith` closers.⁹

The seabed provided. The leading constant *is* the existing 1D quartic partition function lemma—no new closed-form computation needed.¹⁰ The Gaussian tail in `TailBound.lean`¹¹ is the input to which the quartic tail reduces; it is reused here unchanged. Integrability of polynomials against the quartic Boltzmann factor came from the existing `quarticIntegrablePow` (also unchanged).

What was new in this tide. Four theorems in the new `QuarticBoundedPrior.lean` module: a quartic half-line tail bound, a tail-symmetry lemma, a partition-function splitting identity, and the headline bounded-prior bound. None of the four theorems introduces a new mathematical

⁸Branches: `tide/2d-pure-quartic` and `tide/sextic-moments`.

⁹Specifically: `integrableExpNegMulSq`, `setIntegralMonoOn`, and `intervalIntegral.integralIicAddIoi`.

¹⁰`Laplace.OneD.quarticPartition` from `Quartic.lean`.

¹¹`gaussianTailBoundRescaledIoi`, originally written for the harmonic case and used in `Localisation.lean`.

technique; they wire existing seabed and Mathlib pieces in a new arrangement. The honest line-count breakdown is roughly: 5 lines of new mathematics (the comparison $w^4 \geq a^2 w^2$ and its consequence), 30 lines of arithmetic manipulation (rewriting $b = ta^2/12$, computing $\frac{1}{b \cdot a} = 12/(ta^3)$, etc.), and 200 lines of integrability and measurability bookkeeping plus splitting/union proofs. That ratio is the dominant cost on a small bounded-prior tide step in this style.

6 Lessons

- *Numerical sanity check pays for itself before writing Lean.* The exponent-constant tightening in the present run came purely from the combination of `scipy.integrate.quad` numbers and a five-minute GPT consult; it cost nothing to find and cost a factor-of-two reduction in the final bound. Worth doing on every tide whose closed form is numerical-friendly. The discipline already lives in the Tide skill; record this run as another data point that confirms it.
- *Concurrent Tide agents need filesystem isolation.* `Laplace.lean` is again the contention point; the situation is identical to the one called out in the 2D pure-quartic retrospective. Until `git worktree` support lands in the Tide skill’s preflight, expect to spend a few minutes per concurrent run on the recovery dance: checkout the import-list file from `main`, re-add only the imports relevant to this tide, and force-retarget the branch label onto the prior tide’s tip before committing. Documenting the dance here should at least make the pattern recognisable on first hit in a future run.
- *The substitution-and-Gaussian-tail template is reusable.* The exact same proof pattern (point-wise comparison $w^{2k} \geq a^{2k-2} \cdot w^2$, then the seabed’s rescaled-Gaussian tail bound) will give a one-line generalisation of the quartic tail lemma to a monomial-of-arbitrary-even-degree tail lemma for $L = w^{2k}/(2k)!$. GPT-5.5 Pro flagged this in the deliberation step (“If you generalise, generalise the tail lemma first, not the full bounded-prior asymptotic theorem”); worth acting on the next time the grammar-precursor tide is taken further.
- *Lean parser is whitespace-sensitive on `/·/`.* For the seabed’s `CLAUDE.md`: when writing absolute-value statements, prefer `|expr| ≤ ...` with no extra outer parens between `expr` and the closing `|`. Lean’s tokeniser treats `) |` as ambiguous in some positions.

7 Follow-ups

- *Bounded-prior quartic with continuous test function* (candidate C from this deliberation). Now the next-step grammar precursor: bounded prior + tail bound infrastructure is in place; what remains is the localisation argument for φ near zero plus the `Asymptotics.IsLittleO` packaging. Estimated 400–600 lines.
- *Bounded-prior monomial generalisation* $\left| \int_{-a}^a e^{-tw^{2k}/(2k)!} dw - \frac{1}{2}((2k)!/t)^{1/(2k)} \Gamma(1/(2k)) \right| \leq C_k(a) \exp(-c_k ta^{2k})$. The tail-bound proof transfers verbatim, using $w^{2k} \geq a^{2k-2} w^2$ for $w \geq a$. The hard part is the `Real.rpow` bookkeeping that comes with the generalised exponent.
- *Bounded-prior harmonic partition* (candidate B). Now trivial given the seabed’s harmonic partition lemma and the splitting idiom from this tide. ~ 50 – 100 lines if anyone wants the literal Gaussian version on the record.

- *Worktree-isolated Tide runs.* Repeated from the 2D pure-quartic retrospective. The recurrence here makes the case stronger.
- *Promote the splitting/union idiom to lean-common.* The hand-written $\text{Iio}(-a) \cup \text{Icc}(-a,a) = \text{Iica}$ disjointness + union proof is a self-contained, generic statement that will recur on every bounded-prior tide. Six lines is small enough to duplicate once but it should not duplicate three times.

Acknowledgements

GPT-5.5 Pro contributed the deliberation-stage advisory at Step 2 and the strategic move that became the final proof strategy: the quartic-tail sublemma via the comparison $w^4 \geq a^2 w^2$. The full consult and the tide log itself (with candidate drafts, vote, numerical check, and result) are preserved alongside one another in the grammar project's tide-log directory under the SRI repo.